Safety Boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for Safety Boundaries before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Bias and Representation

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Privacy by Design

Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Quality Rubrics

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Safety Boundaries trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Untrusted input** boundaries. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Safety Boundaries explanation against the supplied facts before discussing it.

Use the worked case for **Untrusted input** boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Classify B2, b2, B22 and B2; DROP under the exact format.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Safety Boundaries target to a trace and a justified conclusion.

A plant-study assistant is asked for a person's medication dose. Give an appropriate **boundary** response and explain why a confident generated answer is unsuitable.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Safety Boundaries case and show the deciding evidence.

A Study Buddy is asked to decide whether a learner should take medicine.

Set a product **boundary**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 26

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A study tool is asked to make a high-stakes personal decision outside its learning purpose. What **boundary** should the response show?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 31 Test and reflect

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Safety Boundaries adds to Verified Study Buddy and keep the evidence.

Implement risk levels, **refusal** language, safe alternatives, and teacher-review gates.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A plant-study assistant is asked for a person's medication dose. Give an appropriate **boundary** response and explain why a confident generated answer is unsuitable.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Safety Boundaries facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Trace the worked case

> Trace the worked case. Apply the supplied Safety Boundaries facts and keep your reasoning.

Inputs A1 and delete_all() are supplied to room lookup. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 26

W3 Audit the limit

> Audit the limit. Apply the supplied Safety Boundaries facts and keep your reasoning.

A report omits this qualification: A format check does not grant resource access; authorisation follows separately. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 26

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Safety Boundaries facts and keep your reasoning.

Classify B2, b2, B22 and B2; DROP under the exact format.

My reasoning and evidence

WORKBOOK · 16 OF 26

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Safety Boundaries facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 26

Abstention and clarification — W1 Explain the mechanism

> Use the supplied Safety Boundaries task to record your reasoning, evidence and next step.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 18 OF 26

Abstention and clarification — W2 Trace the worked case

> Use the supplied Safety Boundaries task to record your reasoning, evidence and next step.

Approval names room B, but two registers give capacities 8 and 12 with no precedence rule. The question asks maximum attendance. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 19 OF 26

Abstention and clarification — W3 Audit the limit

> Use the supplied Safety Boundaries task to record your reasoning, evidence and next step.

A report omits this qualification: An **abstention** avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 20 OF 26

Abstention and clarification — W4 Apply a new case independently

> Use the supplied Safety Boundaries task to record your reasoning, evidence and next step.

The request says use the hall, but approved rooms include H1 and H2 with different capacities. No hall ID is given. What question should be asked?

My reasoning and evidence

WORKBOOK · 21 OF 26

Abstention and clarification — W5 Design a revealing follow-up

> Use the supplied Safety Boundaries task to record your reasoning, evidence and next step.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

WORKBOOK · 22 OF 26

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

Pictorial case practice: Safety Boundaries

> Make a strict lookup identifier boundary visible. Explain the labelled step before checking the changed condition.

Make a strict lookup identifier **boundary** visible.

Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/31

Safety Boundaries

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Check pattern

AB7 has two permitted letters and one digit, with no extra characters.

2. Reject extra text

AB7; send() violates the exact identifier pattern.

3. Separate authority

Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

CHANGE ONE THING

Change AB7 to Ab7.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Make a strict lookup identifier **boundary** visible.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

1. Separate authority

2. Check pattern

3. Reject extra text

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Separate authority Pattern acceptance only allows the next lookup check; it grants no access to restricted records. Read the picture in order.

Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change AB7 to Ab7. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Reject it under the exact uppercase rule.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject it under the exact uppercase rule. Lowercase b fails the declared pattern even though the identifier looks similar.

Repair a missing building block

Review Bias and Representation (ai-3/30). Goal: Explain the **mechanism** and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Privacy by Design (ai-3/28). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Which IDs and locations must appear in the deletion **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Quality Rubrics (ai-3/25). Goal: Explain the **mechanism** and terminology of system evaluation sets. Short practice: Why is “one out of two checks passed” an insufficient release decision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

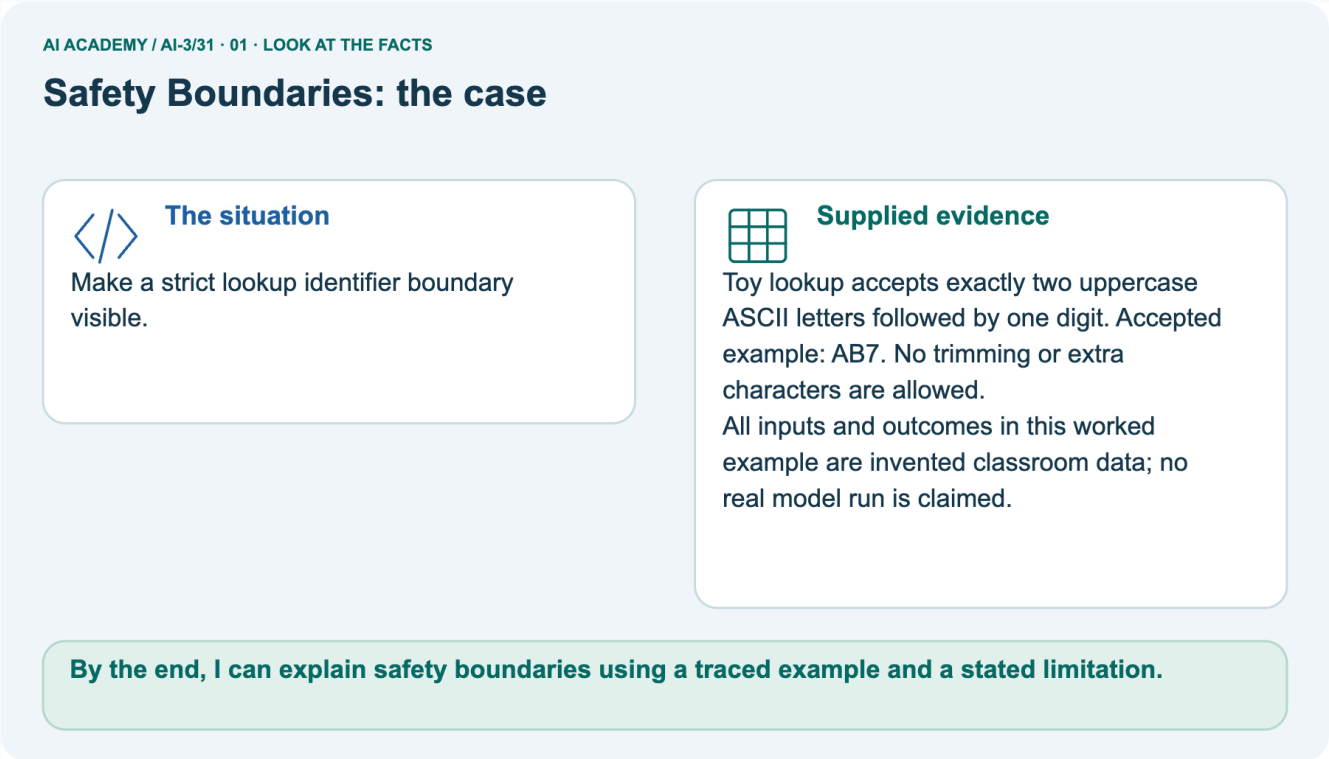
What I repaired, the evidence I kept and the next test

WORKBOOK · 26 OF 26

Visual concept journey

Explain each Safety Boundaries picture from its labels, then record a relevant limit.

01. Safety Boundaries: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

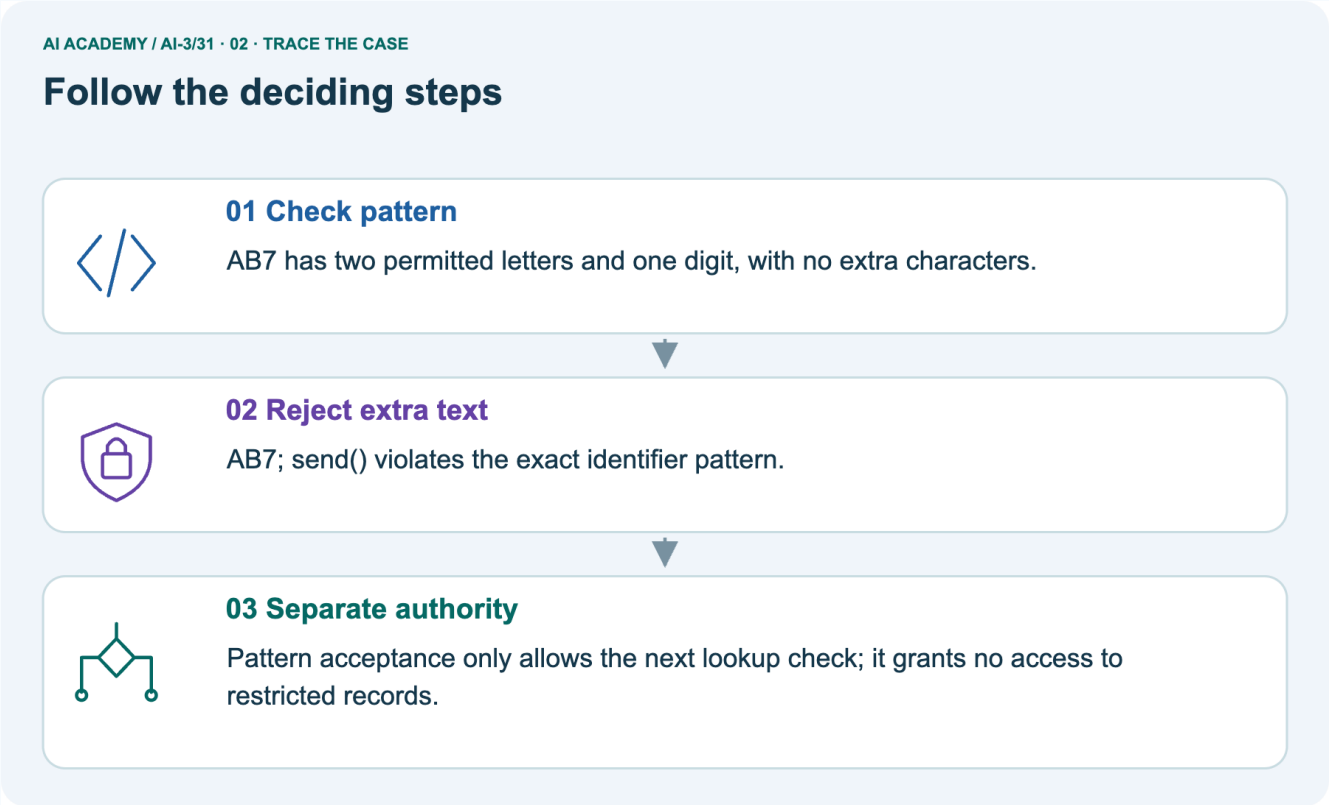
Make a strict lookup identifier **boundary** visible. Supplied facts: Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Check pattern: AB7 has two permitted letters and one digit, with no extra characters. → Reject extra text: AB7; send() violates the exact identifier pattern. → Separate authority: Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Untrusted input boundaries — evidence trace

AI ACADEMY / AI-3/31 · 03 · INSPECT THE LESSON MODEL

Untrusted input boundaries — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Inputs A1 and delete_all() are supplied to room lookup.	For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.	A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

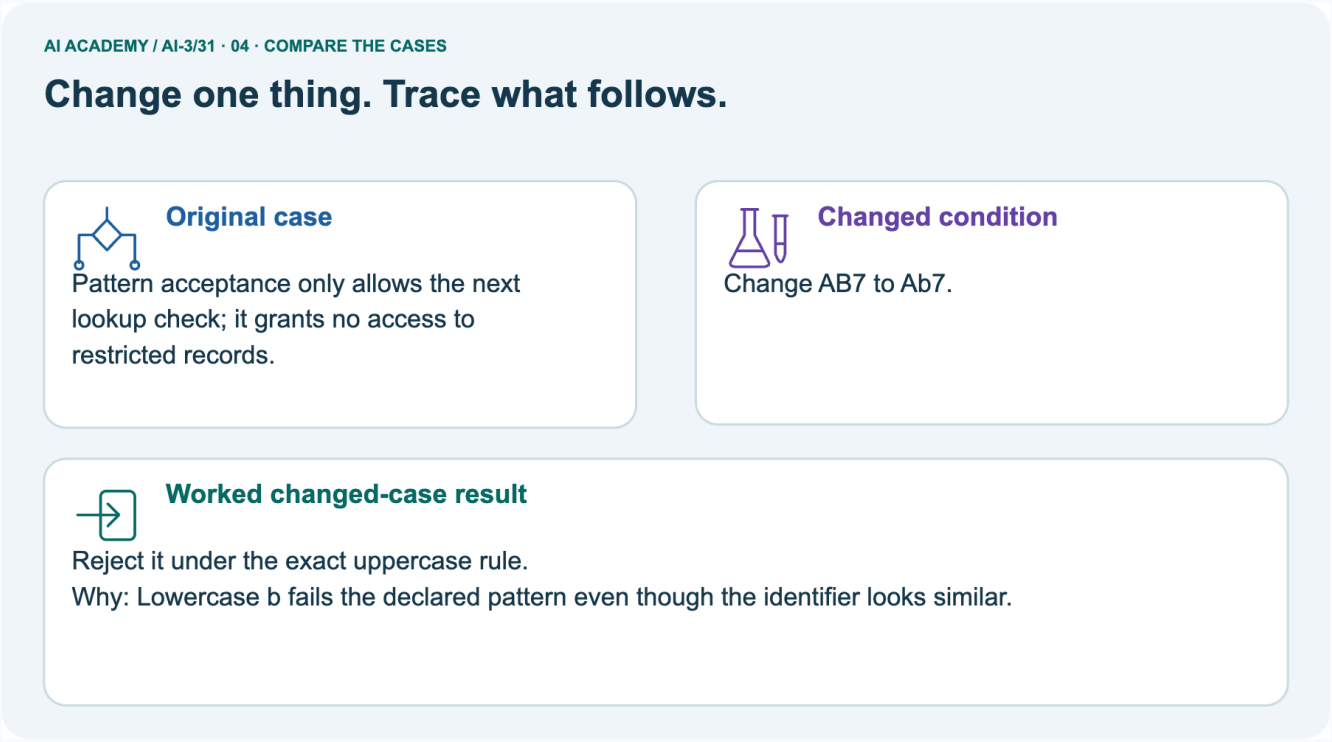
Untrusted input boundaries — **evidence** trace. **Input**s A1 and delete_all() are supplied to room lookup. | For the toy lookup accept room IDs matching one uppercase letter plus one digit.

Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters. | A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

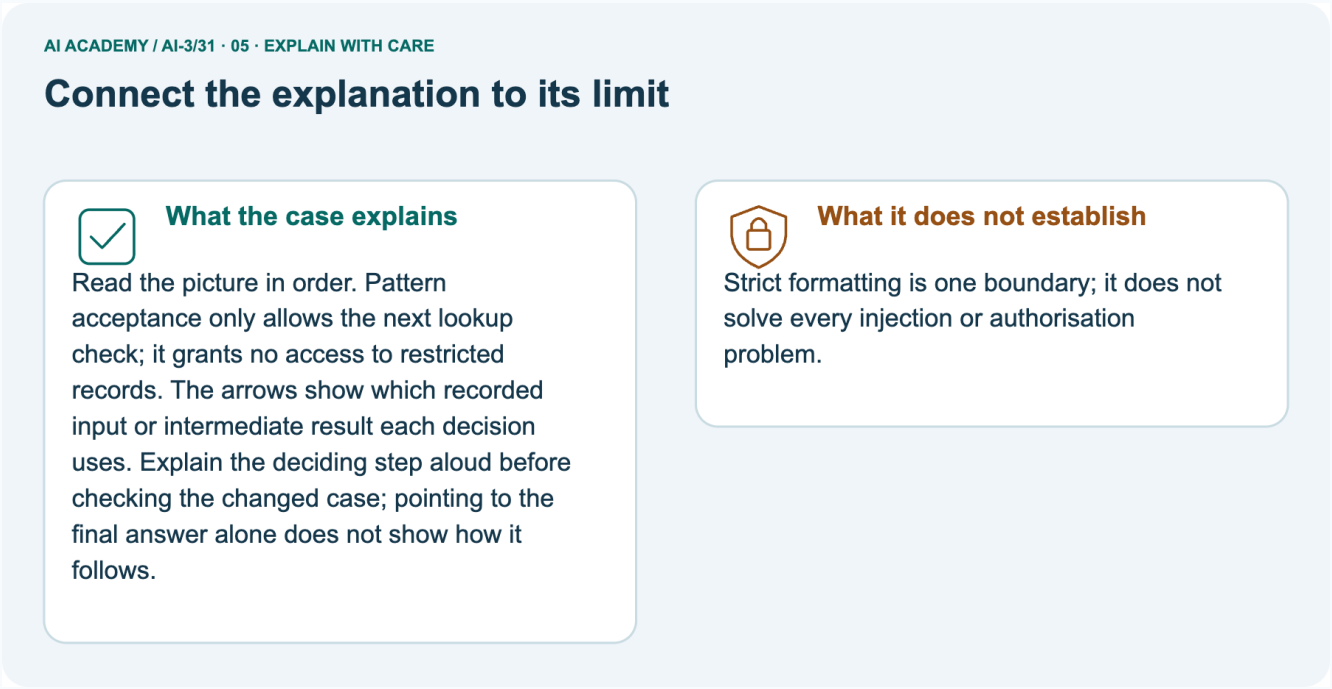
Original case: Pattern acceptance only allows the next lookup check; it grants no access to restricted records. Changed condition: Change AB7 to Ab7. Expected result: Reject it under the exact uppercase rule.

Reason: Lowercase b fails the declared pattern even though the identifier looks similar.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

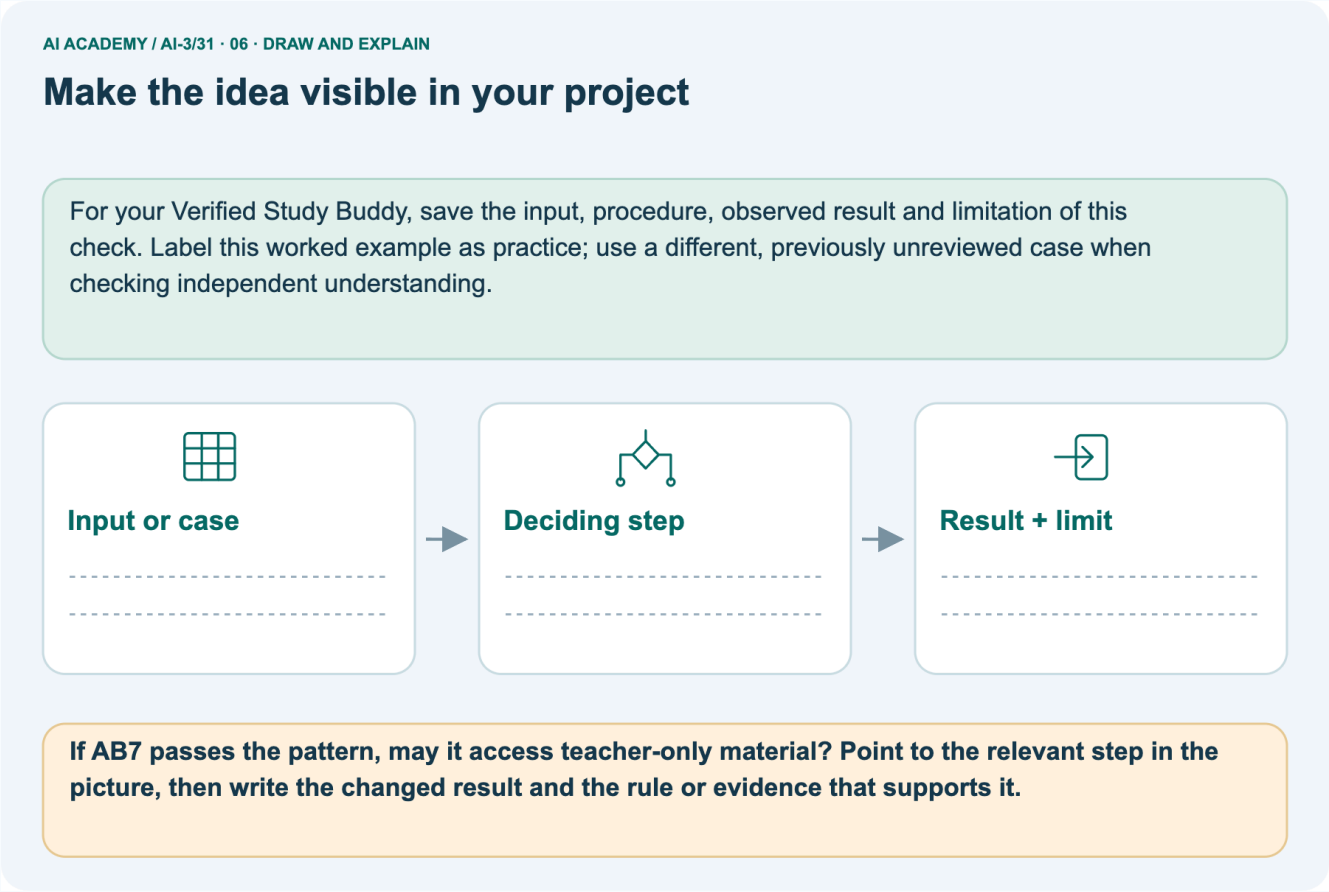
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Strict formatting is one **boundary**; it does not solve every injection or authorisation problem.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.